Ogbonna Arinze

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SUMMARY

Product-minded fullstack engineer with 8 years of experience building and shipping B2B software products end-to-end across healthcare, fintech, and enterprise SaaS. Deep expertise in **TypeScript**, **React**, **Node.js**, **PostgreSQL**, and **Prisma**, with a track record of owning entire product areas from architecture through production deployment. Proven ability to work autonomously, ship rapidly, collaborate directly with customers and stakeholders, and mentor engineering teams while maintaining high code quality through automated testing and CI/CD pipelines.

CORE COMPETENCIES

- **Product & Fullstack Engineering:** End-to-end product ownership, B2B SaaS development, API design, customer-facing feature delivery, and rapid iteration.
- **Languages & Frameworks:** TypeScript, JavaScript (ESNext), Node.js, React, Next.js, NestJS, Vue.js, and Prisma ORM.
- **Backend & Data:** PostgreSQL, MongoDB, REST APIs, GraphQL, JWT authentication, RBAC, SSE streaming, and database schema design.
- **Infrastructure & DevOps:** Docker, Docker Compose, GitHub Actions, CI/CD pipelines, AWS (CloudFront, S3), Vercel, and Nx monorepo orchestration.
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, Storybook, ESLint, Prettier, and CI-driven QA.
- **AI & LLM Systems:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, Mastra, vector search (pgvector, Chroma), and agentic workflows.
- **Leadership:** Technical architecture ownership, mentorship, code reviews, Agile/Scrum, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)
Frontend Engineer — *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, owning the end-to-end product delivery from architecture decisions through production deployment.
- Co-developed a full-stack **Next.js** chat widget powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources and query them in real time via a conversational interface.
- Implemented **JWT-based authentication** and **role-based access control** in **Next.js** with **TypeScript** and **React Query**, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Optimized application performance by implementing lazy loading, code splitting, and CDN asset delivery, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and high code quality standards.

Descasio — Lagos, Nigeria (Hybrid)
Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface for a B2B product, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application, directly improving the end-user experience.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports and more confident releases.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes across the team.
- Mentored junior engineers on frontend best practices and utilized **Storybook** for documenting React components, streamlining change management and supporting agile development.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, processing thousands to millions of daily transactions and boosting settlement rates by **80%**, serving enterprise banking clients.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React**, **Redux**, **RxJS** for real-time data streaming, and **Axios** for API integration.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface in the B2B financial platform.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design**, enabling operations teams to monitor transaction flows.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **Node.js RESTful APIs** and translated business requirements into efficient **MongoDB** database models, delivering end-to-end product features from backend schema design to frontend implementation.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and overseeing task management in a startup environment.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **Node.js**, building and testing **REST API** endpoints for multiple client-facing B2B applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **Node.js** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered platform for Nigerian budget analysis and corruption tracking. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected and shipped the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app (22 interactive chart types with Recharts), Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, including pgvector extension for vector storage, and wrote custom migration tooling that filters out Mastra-managed chunk tables from Prisma's drift detection.
- Built a **multi-agent AI system** using **Mastra** with 5 specialized agents and a three-stage intent classification pipeline, powered by a **hybrid RAG pipeline** combining **pgvector** + BM25 full-text search with Reciprocal Rank Fusion and **Cohere** reranking.
- Implemented **SSE streaming** for real-time chat responses with 15s keepalive heartbeats, phone OTP authentication, **Telegram bot** with webhook handling, and shareable public conversations with SEO optimization (Open Graph, JSON-LD, SSR).
- Configured CI/CD with **GitHub Actions** for automated builds and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and 30s graceful shutdown. Integrated **Playwright** E2E tests across 4 test projects.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*